

Replication of Computerised Assessment of Motion Imitation (CAMI) Study with Neurodivergent Pediatric Population

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Introduction

Motion imitation is fundamental to social learning (Prinz, 2002; Heyes, 2013; Nadel, 2006), with deficits prevalent in autism spectrum disorder (ASD) and linked to social communication impairment (Edwards, 2014; Gowen, 2012; MacNeil, 2012; Nebel, 2016). Traditional imitation assessments involve time-consuming subjective human observation coding (HOC) (Hobson, 2008; Vivanti, 2008; Marsh, 2013). Tunçgenç et al.(2021) proposed an objective and scalable Computerised Assessment of Motion Imitation (CAMI) score, which outperformed HOC in distinguishing ASD from typical development. Santra et al.(2026) found children with ADHD showed similar CAMI performance to neurotypical children. Prior studies suggest that motor alterations are observed in depressive and anxiety disorders, specifically gait parameters and overall abnormal movement (Elkjær, 2022). This abstract replicates the CAMI study in children with ADHD and heterotypic comorbidities, including anxiety and depressive disorders, hypothesising no significant differences from neurotypical children.

Methods

Participants (N=50;27M, 23F; mean age(SD)=10.37(2.40)) were assessed using Healthy Brain Network Phenotypic Assessment Protocol (Alexander, 2017). Participants were recorded using two Kinect cameras while imitating dance movements across three trials (two 1-min sequences, one 1.5min slowed repetition sequence). Stimuli was presented on a large monitor, with cameras mounted in front and behind the participant. Data processing and calculation of CAMI scores was replicated from the study conducted by Tunçgenç et al.(2021). CAMI scores were consistent and significantly correlated across three trials ($p<0.05$). Thus, a composite score, ranging from 0 to 1, was obtained by averaging scores from the three trials, with higher scores indicating better imitation performance.

Results

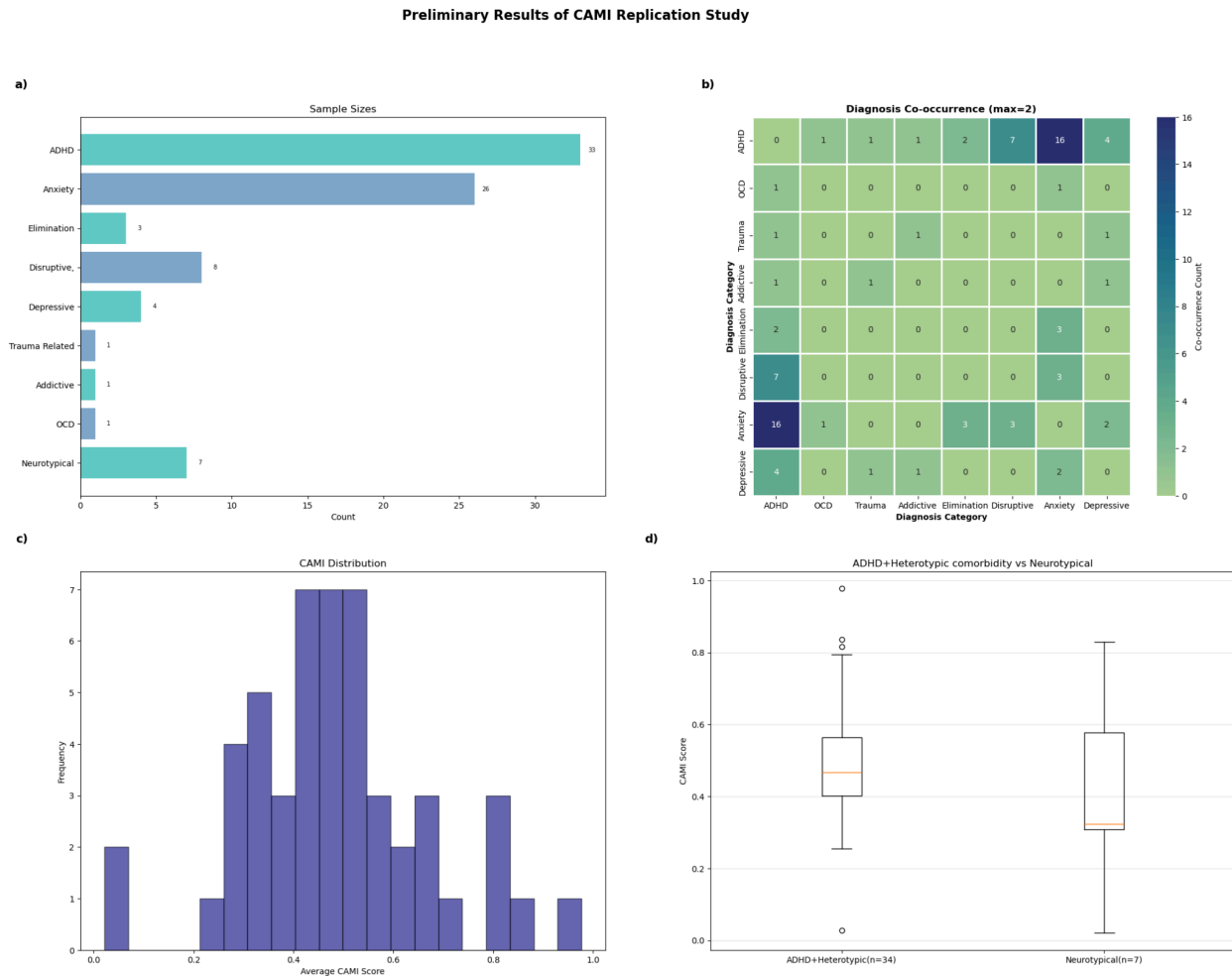
The sample included children with 33 ADHD and heterotypic comorbidities (anxiety, OCD, trauma, addictive, elimination, disruptive and depressive disorders), 10 with anxiety without ADHD, and 7 neurotypical children. Distribution of heterotypic comorbidity is shown in Fig. 1(b). No significant differences in CAMI scores between children with ADHD+ Heterotypic comorbidity ($n=34$) and neurotypical children ($n=7$) ($t(7.39)=0.68$, $p=0.51$) were observed. CAMI score was not correlated to age ($\rho(48)=-0.15$, $p=0.29$) or gender (*male*(SD)= $0.47(0.18)$; *female*(SD)= $0.48(0.19)$). Lower mean CAMI score in neurotypical children (Fig. 1(d)) may be attributed to low sample size.

Discussion

This study showed that motion imitation challenges measured using the CAMI method (Tunçgenç, 2021) are not observed in children with ADHD, regardless of heterotypic comorbidities. As data analysis is ongoing, some future directions include exploring correlation of CAMI scores with ADHD traits evaluated using diagnostic assessments (Alexander, 2017), and investigating effect of comorbidities that demonstrate motor alterations (Elkjær, 2022), further verifying ASD-specificity of CAMI (Santra, 2026). Preliminary findings indicate that children with ADHD and heterotypic comorbidities, and neurotypical children have similar CAMI performance showing that motion imitation impairment is not observed in children with ADHD, even in presence of heterotypic comorbidities. These findings are consistent with Santra et al.(2026) and validate the study by Tunçgenç et al.(2021) as well as the robustness of CAMI methodology across different study sites.

Figure 1

Preliminary results of CAMI Replication Study



Note. This figure shows preliminary results of replicating the CAMI study in neurodivergent pediatric population. (a) A bar plot shows the sample size for each diagnostic group and neurotypical group. Each group count does not add up to total number of participants (n=50) due to high comorbidity. (b) A heatmap shows sample size for multiple co-occurrence diagnostic groups with maximum of two co-occurring diagnoses. (c) Histogram shows the distribution of average CAMI scores across all participants. (d) A boxplot comparing mean CAMI score for children with ADHD+heterotypic comorbidities and neurotypical children.

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